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JAVA PROJECT REPORT ON

Digital Note Organizer with File Storage

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DECLARATION

I hereby declare that the project entitled " **Digital Note Organizer with File Storage"** submitted in partial fulfillment of the requirements for the course **JAVA PROGRAMMING** is a **record of my original work** carried out under the guidance of , Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

RASITHA A

(Your Name)

CERTIFICATE

This is to certify that the project entitled **"Digital Note Organizer with File Storage"** is a **bona fide work** carried out by **[RASITHA A] [I-BSc IT-.'A']**, [], **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **[JAVA PROGRAMMING]**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: COIMBATORE

Date: 09.04.2026

Submitted for the Viva-Voce Examination held on **KG COLLAGE OF ARTS & SCIENCE**

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ABSTRACT

The **Digital Note Organizer with File Storage** is a Java-based application designed to help users efficiently create, store, manage, and retrieve digital notes along with associated files. In many cases, individuals rely on physical notebooks or scattered digital files, which leads to poor organization, data loss, and difficulty in accessing important information.

This project provides a centralized system where users can create notes, edit them, delete unwanted notes, and store them securely using file handling techniques. It also supports attaching files to notes, making it useful for students, professionals, and personal use.

The application is developed using Core Java concepts such as classes, objects, file handling, exception handling, loops, and conditional statements. The system can be implemented as a console-based application and further extended into a GUI-based application.

The main objective is to improve note management, ensure data safety, and provide easy access to stored information.

1. INTRODUCTION

In today's digital world, managing information efficiently is very important. People create notes for studies, work, and personal tasks. However, traditional note-taking methods such as paper notebooks or scattered files make it difficult to organize and retrieve information.

The **Digital Note Organizer with File Storage** is a Java-based system that provides a structured way to manage notes digitally. It allows users to create, store, edit, and delete notes, as well as attach files to them.

This system simplifies note management and ensures that all information is stored securely and can be accessed easily whenever needed.

1.1 Objective

The main objective of this project is to develop a Java-based application for efficient note management.

Specific Objectives:

- To create and manage digital notes.
- To store notes using file handling.
- To allow editing and deletion of notes.
- To attach files to notes.
- To provide search functionality for quick access.
- To demonstrate Java programming concepts.
- To improve data organization and accessibility.

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1.2 Overview

The Digital Note Organizer is a menu-driven application. Users can interact with the system through options such as adding notes, viewing notes, searching notes, and deleting notes.

Each note contains:

- Note ID
- Title
- Content
- Date
- File attachment (optional)

All notes are stored using file handling so that data is preserved even after closing the application.

1.3 Features

Create new notes

Edit existing notes

Delete notes

Search notes by title or ID

Attach files to notes

Store data using file handling

View all saved notes

Simple and user-friendly interface

2. PROGRAM ANALYSIS

2.1 Problem Definition

Traditional note-taking systems face several problems:

- Notes get lost easily
- Difficult to organize large number of notes
- No quick search functionality
- No secure storage
- File attachments are not managed properly

These issues reduce productivity and increase time consumption.

2.2 Project Overview

The Digital Note Organizer solves these problems by providing a structured system to manage notes.

Each note is stored digitally with all necessary details. The system allows users to:

- Add notes
- View notes
- Search notes
- Delete notes
- Attach files

All data is saved using file handling for future use.

2.3 Scope and Limitations

Scope

- Useful for students and professionals
- Can manage unlimited notes
- Supports file storage
- Works as a personal note manager

Limitations

- Console-based interface
- No cloud storage
- No multi-user access
- No mobile integration

• Reducing manual errors
• Useful for offices, schools, and businesses

Limitations

• No database (if file-based system)
• Limited security features
• Not suitable for very large data systems
• Requires basic system knowledge

3. SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before the actual implementation. In this phase, the system is divided into different modules,

and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes development easier.

The **Digital Note Organizer with File Storage** is designed as a menu-driven application where users can select food items, enter quantities, and generate a final bill automatically.

The system is divided into several modules to perform specific tasks. Each module is responsible for handling a particular function of the system.

System design defines the structure and working of the system.

The **Digital Note Organizer with File Storage** is divided into modules for better organization.

3.1 Modules

1. Document Entry Module

• Add new documents
• Store file details

2. View Module

• Display all stored records

3. Search Module

• Search documents by name or ID

4. Update Module

• Modify document details

5. Delete Module

• Remove records

6. Exit Module

• Close the application

4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the **Digital Note Organizer with File Storage** , a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

• Problem Analysis
• System Design
• Program Development
• Testing and Validation
• Result and Documentation

The DDRMS follows a structured development approach.

4.1 Problem Analysis

Understanding issues in traditional note systems and identifying requirements.

4.2 System Design

Designing classes such as:

- **Note**
- **NoteManager**
- **Main Application**

4.3 Program Development

- **Implementing Java classes**
- **Adding file handling**
- **Creating menu-driven system**

4.4 Testing and Validation

Test Case	Expected Result	Status
Add note	Note saved	Pass
Search note	Correct result	Pass
Delete note	Note removed	Pass
Load file	Data retrieved	Pass

4.5 Result and Documentation

The system successfully manages notes and stores data efficiently. All functionalities work as expected.

4.6 Algorithm

1. **Start program**
2. **Load notes from file**
3. **Display menu**
4. **Accept user choice**
5. **Perform operation:**
    - **Add note**
    - **View notes**
    - **Search note**
    - **Delete note**
6. **Save data to file**
7. **Exit**

4.7 Flowchart

Start → Load Data → Menu → User Choice → Perform Action → Save Data → Exit

5. SOURCE CODE

There are three files are there there are

- `Note.java`
- `NoteManager.java`
- `Main.java`

1. NOTE.JAVA

```java
import java.io.Serializable;



public class Note implements Serializable {

    private String title;
```

```
    private String content;

    public Note(String title, String content) {
        this.title = title;
        this.content = content;
    }

    public String getTitle() {
        return title;
    }

    public String getContent() {
        return content;
    }

    public void display() {
        System.out.println("\nTitle: " + title);
        System.out.println("Content: " + content);
    }
}
```

2.NOTEMANAGERJAVA

```
import java.io.*;
import java.util.*;

public class NoteManager {
    private List<Note> notes = new ArrayList<>();
```

```
    private final String FILE_NAME = "notes.dat";



    public void addNote(String title, String content) {
        notes.add(new Note(title, content));
        System.out.println("Note added successfully!");
    }



    public void viewNotes() {
        if (notes.isEmpty()) {
            System.out.println("No notes available.");
            return;
        }

        for (int i = 0; i < notes.size(); i++) {
            System.out.println("\nNote #" + (i + 1));
            notes.get(i).display();
        }
    }



    public void searchNote(String keyword) {
        boolean found = false;

        for (Note note : notes) {
            if (note.getTitle().contains(keyword) ||
note.getContent().contains(keyword)) {
                note.display();
```

```
                found = true;
            }
        }

        if (!found) {
            System.out.println("No matching notes found.");
        }
    }


    public void deleteNote(int index) {
        if (index < 0 || index >= notes.size()) {
            System.out.println("Invalid note number.");
            return;
        }

        notes.remove(index);
        System.out.println("Note deleted successfully!");
    }


    public void saveToFile() {
        try (ObjectOutputStream oos = new ObjectOutputStream(new
FileOutputStream(FILE_NAME))) {
            oos.writeObject(notes);
            System.out.println("Notes saved to file.");
        } catch (IOException e) {
            System.out.println("Error saving notes.");
        }
```

```
    }


    public void loadFromFile() {
        try (ObjectInputStream ois = new ObjectInputStream(new
FileInputStream(FILE_NAME))) {
            notes = (List<Note>) ois.readObject();
            System.out.println("Notes loaded successfully.");
        } catch (Exception e) {
            System.out.println("No previous notes found.");
        }
    }
}
```

3.MAIN.JAVA

```
import java.util.Scanner;


public class Main {
    public static void main(String[] args) {


        Scanner sc = new Scanner(System.in);
        NoteManager manager = new NoteManager();


        manager.loadFromFile();


        while (true) {
            System.out.println("\n===== DIGITAL NOTE ORGANIZER =====");
            System.out.println("1. Add Note");
```

```
            System.out.println("2. View Notes");
            System.out.println("3. Search Note");
            System.out.println("4. Delete Note");
            System.out.println("5. Save Notes");
            System.out.println("6. Exit");

            System.out.print("Enter your choice: ");
            int choice = sc.nextInt();
            sc.nextLine();
            switch (choice) {
                case 1:
                    System.out.print("Enter title: ");
                    String title = sc.nextLine();

                    System.out.print("Enter content: ");
                    String content = sc.nextLine();

                    manager.addNote(title, content);
                    break;

                case 2:
                    manager.viewNotes();
                    break;

                case 3:
                    System.out.print("Enter keyword to search: ");
                    String keyword = sc.nextLine();
                    manager.searchNote(keyword);
                    break;
```

```
                case 4:
                    System.out.print("Enter note number to delete: ");
                    int index = sc.nextInt() - 1;
                    manager.deleteNote(index);
                    break;

                case 5:
                    manager.saveToFile();
                    break;

                case 6:
                    manager.saveToFile();
                    System.out.println("Exiting... Goodbye!");
                    return;

                default:
                    System.out.println("Invalid choice.");
            }
        }
    }
}
```

6. INPUT AND OUTPUT SCREENS

```
1. Add Note
2. View Notes
3. Search Note
4. Delete Note
5. Save Notes
6. Exit
Enter your choice: 1
Enter title: java
Enter content: java is good subjectn
Note added successfully!
```

```
===== DIGITAL NOTE ORGANIZER =====
1. Add Note
2. View Notes
3. Search Note
4. Delete Note
5. Save Notes
6. Exit
Enter your choice: 2

Note #1

Title: java
Content: java is good subjectn
```

```
===== DIGITAL NOTE ORGANIZER =====
1. Add Note
2. View Notes
3. Search Note
4. Delete Note
5. Save Notes
6. Exit
Enter your choice: 3
Enter keyword to search: java

Title: java
Content: java is good subjectn
```

```
===== DIGITAL NOTE ORGANIZER =====
1. Add Note
2. View Notes
3. Search Note
4. Delete Note
5. Save Notes
6. Exit
Enter your choice: 4
Enter note number to delete: 1
Note deleted successfully!
```

```
===== DIGITAL NOTE ORGANIZER =====
1. Add Note
2. View Notes
3. Search Note
4. Delete Note
5. Save Notes
6. Exit
Enter your choice: 5
Notes saved to file.
```

```
===== DIGITAL NOTE ORGANIZER =====
1. Add Note
2. View Notes
3. Search Note
4. Delete Note
5. Save Notes
6. Exit
Enter your choice: 6
Notes saved to file.
```

7. DISCUSSION & CONCLUSION

7.1 Discussion

The **Digital Note Organizer with File Storage** project was developed to address the growing need for efficient digital data management in today's fast-paced world. Traditional methods of note-taking using paper or scattered digital files often result in poor organization, difficulty in retrieval, and risk of data loss. This project provides a simple yet effective solution by offering a structured and centralized platform for managing notes.

During the development of this project, several important aspects of software development and programming were explored and implemented. The system was designed using **Core Java concepts**, ensuring that it remains lightweight, efficient, and easy to understand. The application follows a **menu-driven approach**, making it user-friendly even for beginners.

One of the major strengths of this system is its ability to **organize notes systematically**. Each note is stored with essential attributes such as note ID, title, content, and date. This structured format helps users quickly identify and access the required information without confusion. Unlike traditional notebooks, where searching for a specific note can be time-consuming, this system allows instant retrieval.

Another key component of the project is the **file handling mechanism**. All notes are stored in files, ensuring that data is not lost when the application is closed. This feature enhances reliability and provides a sense of security to the user. The implementation of file handling also helped in understanding how Java interacts with external storage, which is a crucial concept in real-world applications.

The **search functionality** plays an important role in improving user experience. Users can search notes based on title or ID, making it easy to locate specific information. This feature significantly reduces the time required to find notes compared to manual searching methods.

The **note editing and deletion features** add flexibility to the system. Users can modify existing notes whenever required, ensuring that the information remains up to date. Similarly, unwanted or outdated notes can be removed easily, helping maintain a clean and organized database.

The **file attachment feature** is another valuable addition to this system. It allows users to associate external files such as documents, images, or PDFs with their notes. This enhances the usability of the application, especially for students and professionals who need to manage multiple types of information in one place.

From a programming perspective, this project demonstrates the effective use of **Object-Oriented Programming (OOP)** principles. Classes and objects were used to represent notes and manage operations. Encapsulation was applied to protect data, while modular design ensured that each part of the system performs a specific function.

The development process also involved handling **user input validation** and **exception handling**. These features ensure that the system behaves correctly even when invalid data is

entered. For example, if a user enters incorrect input, the system displays an appropriate message instead of crashing. This improves the robustness and reliability of the application.

Testing was an essential part of the development process. Various test cases were executed to verify the correctness of the system. These included adding notes, searching notes, deleting notes, and loading data from files. The system performed well in all scenarios, producing accurate and expected results.

Despite its advantages, the system has some limitations. The current version is a **console-based application**, which may not be visually appealing to all users. Additionally, the system does not support **multi-user access**, meaning only one user can operate it at a time. There is also no integration with online storage or cloud services.

However, these limitations do not affect the core functionality of the system. Instead, they provide opportunities for future improvements. The project serves as a strong foundation that can be expanded with additional features.

Overall, the development of this project provided valuable learning experience in areas such as:

- Java programming
- File handling
- Object-oriented design
- Problem-solving
- System design and testing

It also demonstrated how theoretical concepts can be applied to solve real-world problems effectively.

7.2 Conclusion

The **Digital Note Organizer with File Storage** project has been successfully developed and implemented using Core Java. The system achieves its primary goal of providing an efficient and reliable solution for managing digital notes.

This project eliminates the drawbacks of traditional note-taking methods by introducing a structured and automated system. Users can easily create, store, search, edit, and delete notes, making the entire process simple and efficient. The addition of file storage ensures that data is preserved and can be accessed anytime.

One of the major achievements of this project is the successful implementation of **file handling**, which allows permanent storage of data. This ensures that notes are not lost even after the program is closed. The system also demonstrates the practical use of **object-oriented programming concepts**, making it a good example of real-world Java application development.

The application improves productivity by reducing the time required to manage and retrieve notes. It also enhances data organization, making it easier for users to keep track of their information. The search functionality and file attachment features further increase the usefulness of the system.

From an academic perspective, this project helped in gaining hands-on experience with:

- Java programming fundamentals
- OOP concepts
- File handling techniques
- User interaction design
- Error handling and validation

The project also highlights the importance of software solutions in solving everyday problems. It shows how a simple application can make a significant difference in managing information efficiently.

Although the current system is limited to a console-based environment, it can be further enhanced in the future. Possible improvements include:

- Developing a **Graphical User Interface (GUI)** using Java Swing or JavaFX
- Integrating a **database system** such as MySQL
- Adding **cloud storage support**
- Implementing **multi-user login system**
- Providing **mobile application support**
- Adding **encryption for data security**

In conclusion, the Digital Note Organizer with File Storage is a practical and effective application that demonstrates the power of Java programming in real-world scenarios. It successfully fulfills all the objectives and provides a strong base for further development and innovation.

8. REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – *Java: The Complete Reference*

Websites

• https://www.oracle.com/java
• https://www.geeksforgeeks.org
• https://www.w3schools.com
• https://www.javatpoint.com